

Leaky bucket algorithm

```
#include<stdio.h>

void main()
{
    int in,leakrate,bsize,n,bucket_storage;
    bucket_storage=0;
    printf("Enter the bucket size: ");
    scanf("%d",&bsize);
    printf("Enter the no of input");
    scanf("%d",&n);
    printf("Enter the packet outgoing rate: ");
    scanf("%d",&leakrate);
    printf("Initiall bucket status: contains %d out of %d\n",bucket_storage,bsize);
    while(n!=0)
    {
        printf("\nEnter the incoming packets:");
        scanf("%d",&in);
        printf("Incoming packet size in %d\n",in);
        if(in<=(bsize-bucket_storage))
        {
            bucket_storage=bucket_storage+in;
            printf("Bucket status:contains %d out of %d\n",bucket_storage,bsize);
        }
        else
        {
            printf("Dropped packet %d\n",in-(bsize-bucket_storage));
            bucket_storage=bsize;
            printf("Bucket status: contain %d out of %d\n ",bucket_storage,bsize);
        }
        bucket_storage=bucket_storage-leakrate;
        if(bucket_storage>=0)
```

```
    printf("%d packet tramitted ,bucket status:contains %d packet out of %d\n",leakrate,bucket_storage,bsize);  
    n--;  
}  
if(n==0 && bucket_storage!=0)  
{  
    bucket_storage=0;  
    printf("After sending , Bucket contains %d packet out of %d\n",bucket_storage,bsize);  
}  
}
```